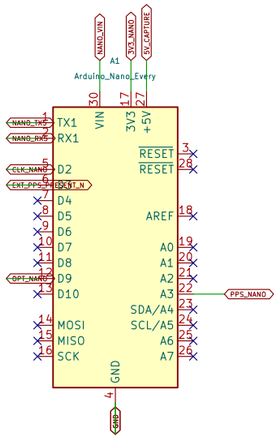
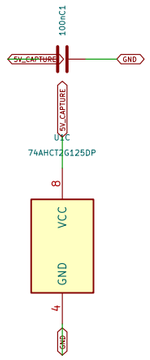
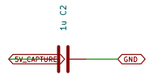
NANO EVERY / SYSTEM

Stock Nano Every and ordinary Micro-USB data cable. USB end faces upper PCB edge.
Nano 5V header is an OUTPUT: carrier external power enters VIN only.
A.9 is an unbuilt prototype. CAD checks do not establish power or timing acceptance.









TimingInputs

File: TimingInputs.kicad_sch

JackInputs

File: JackInputs.kicad_sch

GNSS_UART

File: GNSS_UART.kicad_sch

PowerProtection

File: PowerProtection.kicad_sch

GPS_PPS

File: GPS_PPS.kicad_sch

GNSSPower

File: GNSSPower.kicad_sch

USBPower

File: USBPower.kicad_sch

BoostPower

File: BoostPower.kicad_sch

Unbuilt prototype — review and acceptance required

Pendulum Observatory

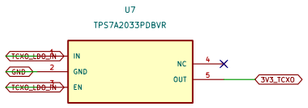
Sheet: /
File: NanoEveryCarrier.kicad_sch

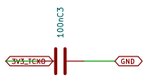
Title: A.9 | NANO EVERY / SYSTEM

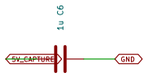
Size: A3	Date: 2026-11-06	**Rev: A.9**
KiCad E.D.A. 10.0.4		Id: 1/9

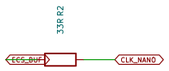
QUIET TCXO / CLOCK

X1 alone uses the dedicated low-noise 3.3V regulator. R1/C16 filter its 5V input.
No GNSS, UART or accessory load may connect to 3V3_TCXO.
Clock output damping and Nano D2 capture clock are retained.

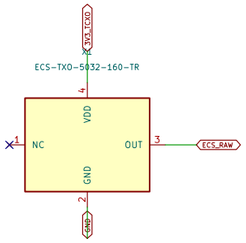


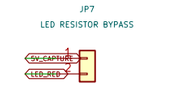




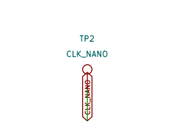


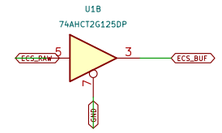














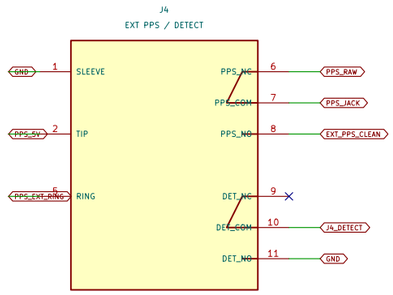
Unbuilt prototype — review and acceptance required
Pendulum Observatory
Sheet: /TimingInputs/
File: TimingInputs.kicad_sch

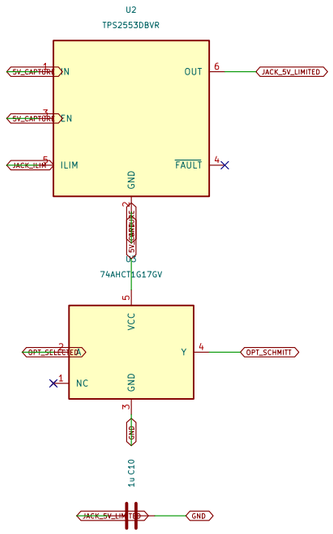
Title: A.9 | QUIET TCXO / CLOCK

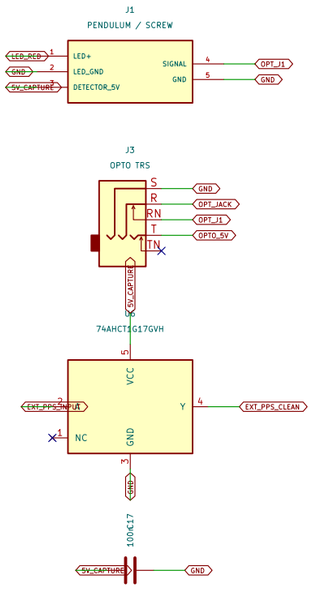
Size: A3	Date: 2026-11-06	**Rev: A.9**
KiCad E.D.A. 10.0.4		Id: 2/9

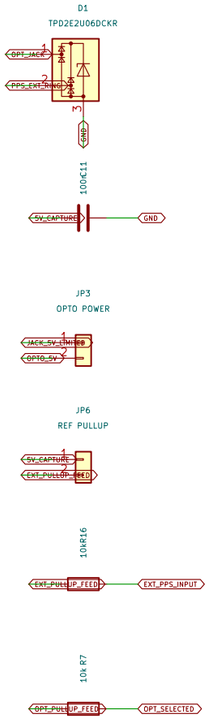
SENSOR / EXTERNAL REFERENCE

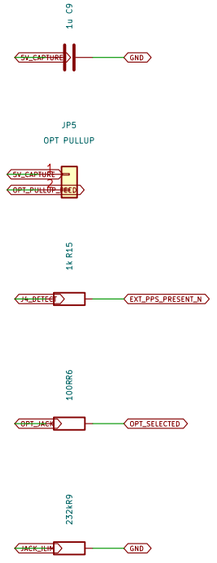
J3/J4: tip = optional5V OUT; ring = signal IN; sleeve = GND. Connect with power off.
J3 insertion selects optical jack; J4 insertion selects external PPS even for an unpowered source.
Pull-ups are selectable. External PPS is conditioned BEFORE isolated J4 source selection.

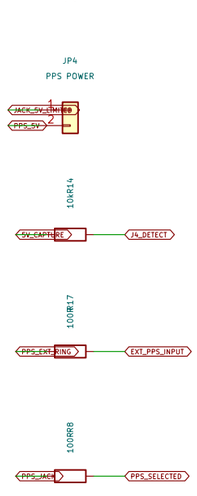












Unbuilt prototype — review and acceptance required

Pendulum Observatory

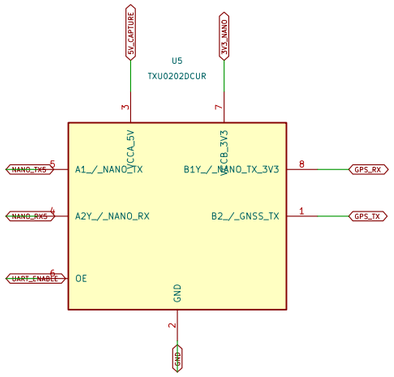
Sheet: /JackInputs/
File: JackInputs.kicad_sch

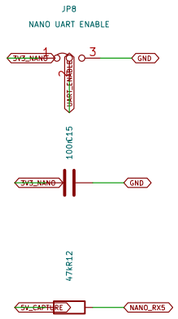
Title: A.9 | SENSOR / EXTERNAL REFERENCE

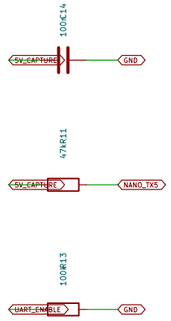
Size: A3	Date: 2026-11-06	**Rev: A.9**
KiCad E.D.A. 10.0.4		Id: 3/9

GNSS UART / SERVICE

Nano Serial1 owns GNSS configuration. USB Serial remains recording/commands.
JP8: 1–2 enabled;2–3 disabled for receiver service. Remove JP2 before separately powering GNSS; UART/PPS can still back-power it.
Hardware support does not implement receiver configuration/qualification firmware.





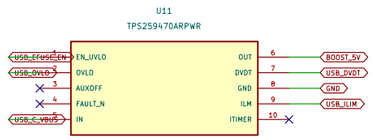


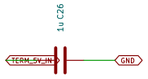
Unbuilt prototype — review and acceptance required
Pendulum Observatory

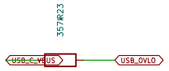
Sheet: /GNSS_UART/		
File: GNSS_UART.kicad_sch		
Title: A.9 | GNSS UART / SERVICE		
Size: A3	Date: 2026-11-06	**Rev: A.9**
KiCad E.D.A. 10.0.4		Id: 4/9

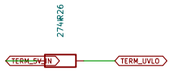
INDEPENDENT INPUT PROTECTION / ORING

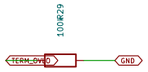
U11 and U12 provide true reverse-current blocking before their common BOOST_5V output.
USB-C has hardware source qualification. Optional J7 accepts regulated4.75–5.25V only.
Nominal protection:1A, OV5.484V; terminal UV turn-on4.488V (falling threshold has IC hysteresis).

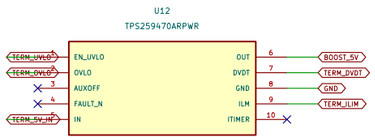




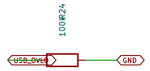


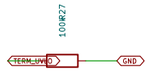


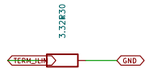


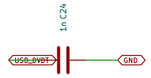


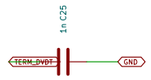


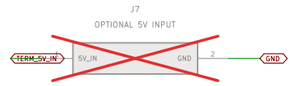


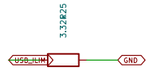


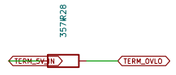












Unbuilt prototype — review and acceptance required
Pendulum Observatory

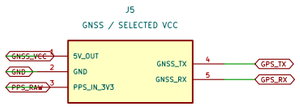
Sheet: /PowerProtection/
File: PowerProtection.kicad_sch

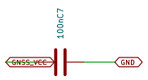
Title: A.9 | INDEPENDENT INPUT PROTECTION / ORING

Size: A3	Date: 2026-11-06	**Rev: A.9**
KiCad E.D.A. 10.0.4		Id: 5/9

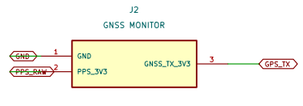
GNSS / PPS SELECTION

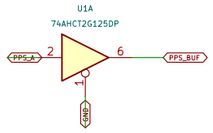
J5: selected3.3V/5V GNSS VCC: nominal3.3V PPS/UART. J2: read-only PPS/TX, no power.
JP9 and JP10 must move together: 2-3 BUFFER default, 1-2 DIRECT comparison.
JP11 normally OPEN: never load boot-shared GNSS PPS without receiver qualification.



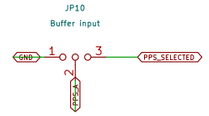


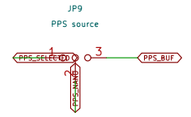


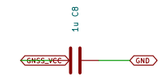
















Unbuilt prototype — review and acceptance required
Pendulum Observatory

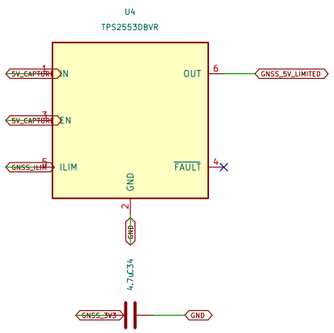
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File: GPS_PPS.kicad_sch

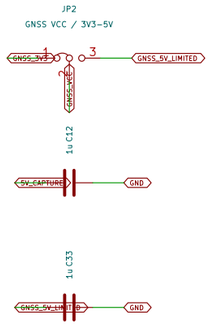
Title: A.9 | GNSS / PPS SELECTION

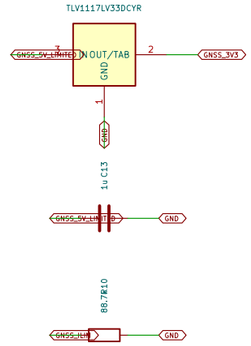
Size: A3	Date: 2026-11-06	**Rev: A.9**
KiCad E.D.A. 10.0.4		Id: 6/9

GNSS POWER / VOLTAGE SELECTOR

U4 protects BOTH5V and3.3V choices;200mA receiver+antenna allowance. U14 dedicated GNSS regulator.
JP2:1–2=3.3V DEFAULT;2–3=5V;removed disconnects J5 VCC. ONE shunt only;all power OFF before changing.
U14 tab is OUTPUT;no reverse-current protection. Independently powered receiver requires JP2 removed.





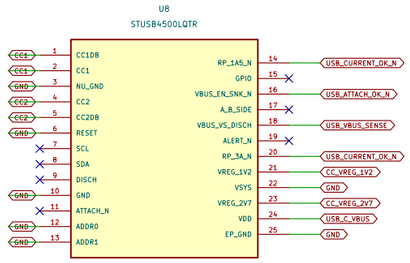


Unbuilt prototype — review and acceptance required
Pendulum Observatory

Sheet: /GNSSPower/		
File: GNSSPower.kicad_sch		
Title: A.9 | GNSS POWER / VOLTAGE SELECTOR		
Size: A3	Date: 2026-11-06	**Rev: A.9**
KiCad E.D.A. 10.0.4		Id: 7/9

USB-C 5V SINK / HARDWARE QUALIFICATION

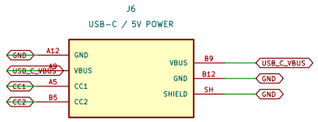
U8: factory-default standalone5V sink; CC1DB/CC2DB connected. No PD and no Nano firmware needed.
U10 enables U11 only when attached AND source advertises1.5A or3A. Default-current sources are rejected.
U9 powers gate logic separately. U8 internal regulator pins are DECOUPLING ONLY.

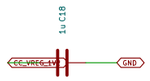


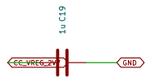


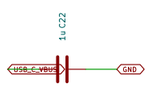






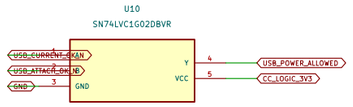


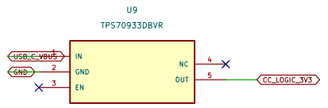


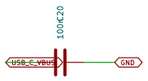


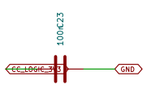














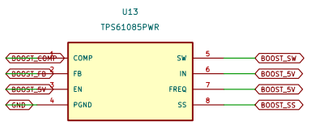
Unbuilt prototype — review and acceptance required
Pendulum Observatory
Sheet: /USBPower/
File: USBPower.kicad_sch

Title: A.9 | USB-C 5V SINK / HARDWARE QUALIFICATION

Size: A3	Date: 2026-11-06	**Rev: A.9**
KiCad E.D.A. 10.0.4		Id: 8/9

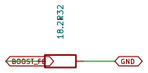
EXTERNAL5V TO NANO VIN

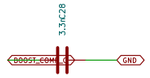
Boost8.04V: D3 blocks any Nano VIN voltage from feeding the boost capacitors or converter.
Nano USB powers its5V rail through Nano D2, separately from the VIN buck input.
JP1 disconnects external VIN feed only. USB load/rail voltage and transfer continuity need bench acceptance.

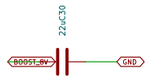


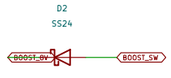


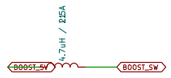


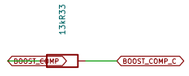


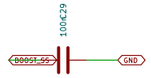


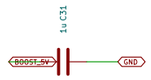


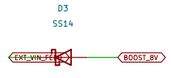


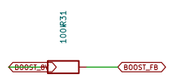












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Title: A.9 | EXTERNAL5V TO NANO VIN

Size: A3	Date: 2026-11-06	**Rev: A.9**
KiCad E.D.A. 10.0.4		Id: 9/9